

GLHS: A Co-Versioned Disclosure-to-Commit Governance Contract for Longitudinal Health AI

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Abstract

Stateful health AI systems face a consistency problem that is not solved by retrieval quality alone: an AI may read an authorized longitudinal-state snapshot and return a persistent proposal after the underlying record, consent state, authorization, or policy has changed. We study this read-to-write interval as a systems-governance problem. GLHS defines a co-versioned contract that binds a task-bounded health-state snapshot (THSS) to a later persistent proposal and revalidates both state and governance before commit through a governed state transition (GST). The implementation preserves bitemporal state, provenance, disclosure scope, snapshot identity, digests, policy and consent versions, and optimistic-concurrency coordinates, while explicitly separating established mechanisms from the cross-operation binding evaluated here. We evaluate three distinct properties: deterministic contract conformance, PostgreSQL state-version concurrency and service-layer overhead, and context utility in a frozen 64-subject synthetic two-model cohort. Snapshot-bound proposals failed closed on the tested identity, digest, evidence, state, policy, consent, purpose, role, task, and expiry mismatches. PostgreSQL races prevented stale commits in the tested state-version path, while profile-global versioning produced the expected false-stale cost for unrelated writes. Strict THSS achieved 63/64 all-axes exact subjects with Claude and 64/64 with Gemini; six of ten prespecified paired contrasts remained Holm-significant, but only 9–21 discordant subjects informed those significant tests. The evidence supports a traceable software path from governed disclosure to persistent write under the tested conditions; it does not establish clinical effectiveness, governance-race atomicity, downgrade prevention for every model-derived path, or regulatory compliance.

Keywords: longitudinal health AI; stateful AI; governed memory; bitemporal state; snapshot binding; consent; optimistic concurrency; transactional agent memory

1 Introduction

Longitudinal health records are difficult for reasons that go beyond retrieval. A fact may be clinically valid at one time and only become known later. Another source may disagree with it. The same information may be appropriate for one actor or purpose but not another. A system can therefore retrieve something relevant and still show an AI the wrong historical state, or more of the record than the current use permits. The difficulty continues after inference: the record, consent state, authorization, or policy may change before a model-derived proposal is written back.

The components needed to handle these problems are established. Medical database work separates valid time from transaction or knowledge time [1, 2, 3]. FHIR Provenance and Consent represent

lineage and policy-relevant information, and FHIR HTTP supports version-aware updates [4, 5, 6]. openEHR maintains versioned EHR content and reconstructable contributions [7], while W3C PROV provides a general provenance model [8]. Recent longitudinal-health systems preserve temporal conflicts, reconcile patient/EHR streams, and maintain longitudinal memory [9, 10, 11, 12, 16]. Agent-memory and tool-runtime work now also provides transactional commit, source-supported updates, recovery, scoped propagation, bitemporal write contracts, and staged irreversible effects [13, 14, 15, 22, 21]. Concurrency and authorization work has moved even closer to the read–write interval: verified agent runtimes formalize read–generate–write anomalies [23], commit-time authorization requires authority witnesses to remain valid at durability [24], and policy-state serializability rechecks stateful authorization against policy state at commit [25].

These results narrow the unresolved question. GLHS does *not* claim that transactional commit, commit-time authorization, bitemporal concurrency control, or governance revalidation is new in general. The candidate gap is health-specific and disclosure-conditioned: *can the exact purpose-, consent-, actor-, and task-bounded health disclosure actually consumed during inference remain an auditable commitment dependency for a later persistent proposal?* A base-version or generic authority check can establish freshness or eligibility without necessarily identifying the health content and governance coordinates that supplied the model’s evidence. Conversely, read-time minimization can constrain model context yet disappear from the later write-admission path.

This work contributes four things without claiming novelty for bitemporality, provenance, consent, conflict preservation, task-bounded disclosure, optimistic concurrency, transactional memory, or commit-time authorization individually:

- a health-specific **disclosure-conditioned, co-versioned commit contract** that carries the persisted identity of the actual governed disclosure and its state/governance coordinates into a later model-derived persistent proposal;
- a reference implementation in CLARA-Care that validates snapshot identity, digest, declared evidence, state version, policy and consent versions, actor/role/purpose/task coordinates, and expiry before persistence;
- an evaluation that separates **contract conformance, database concurrency/overhead**, and **LLM context utility** instead of pooling them into one superiority claim; and
- explicit negative boundaries: provenance-sensitive downgrade prevention is not mandatory on every model-derived path, governance-writer TOCTOU atomicity is not demonstrated, profile-global versioning creates false stale rejection, and clinical correctness is not independently adjudicated.

This arXiv line is systems-oriented: future versions are intended to preserve frozen artifact identifiers and clearly separate newly rerun evidence from prior results rather than retroactively reassigning old runs to new implementations.

2 Related Systems and Design Gap

The 2026 literature substantially narrows the defensible novelty boundary. MemTX and MemTxn already make persistent agent-memory updates transactional and evidence-aware [13, 14]; Cordon stages and validates irreversible effects at a task-level semantic transaction boundary [21]; TOKI formalizes bitemporal write-time correctness and provenance [22]; and verified agent runtimes make read–generate–write concurrency anomalies explicit [23]. More directly, CommitGuard studies stale authority at the durability boundary [24], while Provenact defines policy-state serializability for concurrent agentic effects [25]. GateMem and governed shared-memory work evaluate access-control and propagation failures

in multi-principal memory [26, 15]. In health AI, bitemporal state arbitration, discrepancy reconciliation, longitudinal memory, and privacy-aware context selection are also active research areas [9, 11, 16].

Accordingly, GLHS is positioned neither as a new transaction mechanism nor as a generic commit-time authorization scheme. The narrower target is the *disclosed health-state object itself*: a persisted purpose/consent/actor/task-bounded disclosure that the AI actually consumed is carried forward as an auditable dependency of the proposed write. We did not identify prior work in this review that evaluates this exact health disclosure-to-persistent-proposal dependency while jointly retaining its state and governance coordinates at write admission. Table 1 states this boundary conservatively.

Table 1: Nearest mechanisms and the narrowed GLHS novelty boundary.

Work / line	Already established	Boundary retained for GLHS
FHIR + openEHR [4, 5, 6, 7]	Provenance/consent representation, record versioning/history, conditional/version-aware updates.	Standards substrate; not itself an evaluated model-consumed disclosure-to-write dependency.
Longitudinal health state [9, 11, 16]	Bitemporal arbitration, reconciliation, longitudinal/private memory, privacy-aware selection.	GLHS does not claim temporal state or private health memory as new.
MemTX / MemTxn [13, 14]	Transactional/evidence-supported memory commit, source checking, recovery, validity/provenance.	GLHS does not claim transactional memory; it binds one health disclosure instance to a later proposal.
Cordon / TOKI [21, 22]	Semantic transactions for effects; bitemporal write correctness and audit/provenance.	GLHS is narrower: what was disclosed to the model remains part of later write admission.
Verified concurrency [23]	Machine-checked read-generate-write anomaly hierarchy and prevention runtimes.	Concurrency control is not novelty; GLHS adds health disclosure/governance dependency semantics.
Commit-time authorization / Provenact [24, 25]	Fresh authority at durability; policy-state serializability for concurrent effects.	Closest general authorization neighbors. GLHS additionally binds the exact governed health disclosure consumed by inference, including purpose/consent/content identity.
GateMem / governed shared memory [26, 15]	Multi-principal access control, forgetting, scoped propagation, provenance.	Evaluates memory governance broadly; GLHS follows a disclosed snapshot into persistent write admission.
GLHS	THSS disclosure record, bound proposal, GST revalidation.	Claim is limited to disclosure-conditioned health read-write governance under the tested implementation/evidence.

3 System Model and Governance Contract

3.1 Governed longitudinal state

GLHS represents a system view that is evidence-backed and permitted for the current use, not an unquestionable biological truth. Evidence, events, assertions, state, transitions, conflicts, and projections remain separate so that provenance and uncertainty are not flattened into the current view. State can be reconstructed at clinical valid time t_v and semantic knowledge time t_k ; database write time remains in the audit trail rather than substituting for knowledge time.

For subject u , the governed state is

$$S_u(t_v, t_k) = \text{GovernedView}_u(t_v, t_k). \quad (1)$$

Assertions retain provenance, epistemic status, and lifecycle. Conflicting assertions may remain visible together until a resolution becomes both valid and known.

3.2 Task-Bounded Health State Snapshot

THSS is the governed object produced on the read path. Besides selected content, it stores the coordinates needed to reproduce what the AI was allowed to see:

$$H = \langle u, a, r, \phi, \tau, v_s, v_\pi, v_c, t_v, t_k, id_H, d_H, E_H \rangle. \quad (2)$$

Here a and r identify actor and role; ϕ and τ denote purpose and task; v_s , v_π , and v_c are state, policy, and consent/governance versions; id_H names the persisted snapshot; d_H is its consistency digest; and E_H is the disclosed content.

Canonical bytes use the project-defined `clara.canonical-json.v1` profile and SHA-256. The profile sorts object keys, preserves array order, emits compact UTF-8 JSON, normalizes timezone-aware datetimes to UTC and negative zero to zero, and rejects unsupported or non-finite values; it is not presented as RFC 8785. The persisted manifest records `glhs.snapshot.v3`, payload schema `glhs.snapshot.payload.v3`, the digest algorithm and canonicalization profile, and validation fails closed when metadata or recomputed digests disagree. Strict model-facing disclosure additionally records consent basis, expiry, assertion identifiers/hashes, evidence identifiers, and an opaque profile scope.

3.3 Snapshot-bound proposal and commit invariant

The model is not given unrestricted write access. It returns a persistent proposal

$$P = \langle u, a, r, \phi, \tau, v_s, v_\pi, v_c, id_H, d_H, \Delta, \rho \rangle, \quad (3)$$

where Δ is the proposed change and ρ records proposal provenance, including a model-manifest reference when applicable. For the snapshot-bound mode studied here,

$$\text{Commit}(P) \Leftrightarrow \text{Bound}(P, H) \wedge \text{StateCurrent}(P) \wedge \text{GovernanceCurrent}(P) \wedge \text{SnapshotValid}(P). \quad (4)$$

Bound fails when profile, actor, role, purpose, task, base state version, snapshot identity/digest, or declared evidence set disagree. At commit, the gateway obtains a **FOR UPDATE** lock on the profile row, rechecks base state, rereads current consent, compares the frozen policy version, and writes the transition and resulting state in the same database session.

The current gateway also supports a base-version-only path. At the manuscript freeze, it does *not* enforce a provenance-sensitive policy that forces every model proposal derived from Strict THSS to remain snapshot-bound after review. Therefore Eq. 4 characterizes the stronger path that is evaluated; the work does not claim general downgrade prevention.

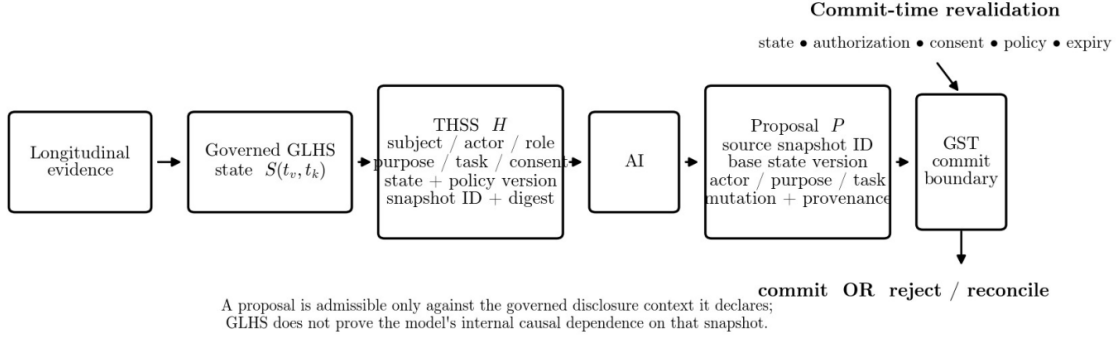


Figure 1: Core GLHS disclosure-to-commit contract. THSS constrains the model-facing disclosure; the proposal carries its source coordinates; GST commits only after the tested binding, state-current, governance-current, and snapshot-valid checks succeed.

3.4 Failure model and claim boundary

Table 2 makes explicit which failure modes are exercised and which remain open.

Table 2: Failure modes, intended handling, and current evidence boundary.

Failure mode	Contract handling	Evidence / limitation
Stale base state	StateCurrent + profile-row lock	Exercised on PostgreSQL state-version races; stale writes rejected in tested path.
Snapshot substitution or digest mismatch	Persisted id_H , d_H and recomputation	Deterministic conformance tests fail closed on tested mismatches.
Actor/role/purpose/task mismatch	Exact coordinate equality in Bound	Exercised in clause-level tests.
Evidence-set substitution	Declared evidence membership in Bound	Exercised in deterministic conformance path.
Consent/policy change during inference	Commit-time reread/version comparison	Current checks exist, but concurrent governance-writer serialization/TOCTOU race atomicity is not demonstrated.
Expired disclosure	SnapshotValid	Expiry is included in revalidation/conformance tests.
Downgrade to base-version-only proposal	Requires provenance-sensitive mandatory policy	Not prevented for every model-derived path at the manuscript freeze.
Unrelated concurrent writes	Profile-global optimistic versioning	Fails closed but creates expected false-stale rejection; finer-grained alternatives were not run on PostgreSQL.

4 Implementation and Reproducibility Boundary

The reference implementation is part of CLARA-Care and is frozen for this manuscript at Git revision 2960acb590fda7ee914b6caded961c89f5e3b8e4. Snapshot-bound proposals are validated in the API-owned commitment gateway. Strict disclosures read persisted manifest identity and digests after canonicalization and fail if required binding fields are missing. State replay exposes separate valid-time

and semantic knowledge-time cutoffs. Historical conflict visibility is reconstructed from conflict-creation and conflict-resolution transitions rather than from mutable current status. Generic and commitment THSS share the same executable five-stage trace. Each reported run keeps its own implementation SHA and checksum lineage; older results are not reassigned to this manuscript-level freeze.

Model-mediated evaluation used the completed 64-subject solver-v5 cohort on 11 August 2026. The run was tied to implementation 17dd4b8c558c2ec0cb8c2572728093d9aa3ce914, internal pre-registration SHA-256 99a6c1338af87e549fa05406c2c7248713d39fa297e56494fb5bd099efdd6bb2, and pre-execution seal 3280f9c9863a76b7bd51891f9bac28401d652d9ce714f6a340099fa37efdb456. Provider calls used an OpenAI-compatible /chat/completions endpoint with requested IDs `antigravity/claude-sonnet-4-6` and `antigravity/gemini-3.6-flash-high`; returned IDs had to match `claude-sonnet-4-6` and `gemini-3.6-flash-high`. Requests used temperature 0, `max_tokens=1024`, `stream=false`, and JSON-schema response mode; `top_p` and a provider API-version parameter were not set. Timeout was 60 s with at most two retries for declared transient HTTP failures and 0.25-s initial backoff. The completed run had zero errors and zero retries.

Service-layer measurements were produced at implementation 36642787931e5ce429f73e8087c6a2ef66e71307 with checksum inventory 75115757d7acec428d9291c567e12a2477f909990627d3f10e8bce40a8911336. Source code and reproducibility materials are available at <https://github.com/Project-CLARA-HBT/CLARA-Care>.

5 Evaluation Design

5.1 Contract clause isolation

The contract experiments and model comparators answer different questions. The standards-composed mechanism comparator includes bitemporal resolution, version-aware writes, current authorization, provenance, and audit. No single comparator was verified to match GLHS on every current consent, policy, purpose, and expiry semantic while differing only in exact snapshot binding, so it is a mechanism comparator rather than a binding-only causal ablation.

Seven progressively stronger variants were applied to the same 16 developer-authored cases, yielding 112 deterministic conformance executions, not 112 independent samples. Variants add: temporal/provenance resolution; base-version checking; authorization at disclosure; provenance and audit; snapshot identity; full context/digest plus disclosed-evidence membership; and current reauthorization with policy, consent, and expiry checks. Separate property-based, adversarial, leakage, and boundary tests are not pooled with the 112 cells; mutation testing was not part of this study.

5.2 PostgreSQL concurrency and service overhead

State-version concurrency was exercised on PostgreSQL 16.14. A four-writer test covered both a shared semantic slot and four unrelated slots under a profile-global version counter. A second grid used 1, 2, 4, 8, and 16 writers, with five independent profile races per level, for 50 races and 310 writer attempts. Losses on the same semantic dependency were classified as true stale; rejections on unrelated slots were classified as false stale. These runs did not include simultaneous authorization loss, consent revocation, or policy changes and therefore do not test governance TOCTOU races.

Service-layer overhead used a clean, one-worker, in-process PostgreSQL 16.14 run with read-committed isolation, synchronous commit, history depth 50, and 30 repetitions per operation. HTTP transport, worker scaling, and provider inference were excluded; p99 is omitted because 30 repetitions do not

support a stable tail estimate.

5.3 Frozen two-model context-utility cohort

Strict THSS was compared with full authorized history, long chronological context, Naive RAG, last-write-wins (LWW), and BTSA. These contrasts test disclosure context for model reasoning, not the later GST persistence boundary. The preregistered endpoint was subject-level all-axes exact match. Ten paired tests (five comparators for each model) were analyzed in one Holm family. Missing or malformed outputs counted as errors. Paired subject-level differences were bootstrapped 1,000 times with seed 20260811; exact two-sided sign tests used only discordant subjects and were Holm-adjusted across all ten tests. The 64-subject plan targeted at least 51 non-tied pairs and approximately 81.5% power for directional win probability 0.75 under a conservative 0.005 per-test threshold; observed significant contrasts ultimately contained only 9–21 non-tied subjects.

6 Results

6.1 Contract conformance

The focused tests exercised Eqs. 2–4. Snapshot-bound proposals were rejected for mismatches in profile, actor, role, purpose, task, base state version, snapshot identity, or digest. The write path also rechecked current policy, consent, and expiry. Incomplete binding metadata and THSS traces with missing or reordered stages were rejected. A historical regression confirmed that a conflict resolved in August still appears in a July snapshot compiled later because conflict visibility is replayed from transition history rather than read from the mutable current projection.

All 112 cells in the 16-case $\times$ 7-variant matrix produced the expected conformance result and identified the first rejecting clause. Because the standards-composed comparator is not matched on every current governance semantic, this matrix is not a causal estimate of exact snapshot binding alone.

6.2 Atomic write safety and version granularity

In the $N = 4$ PostgreSQL test, both same-slot and unrelated-slot races produced one atomic winner and three stale rejections with no database error. Under these tested conditions the profile-global contract failed closed. The contention grid reproduced the expected consequence of one winner per simultaneous profile race: unrelated-slot false-stale rejection was 0 with one writer, 0.50 with two, 0.75 with four, 0.875 with eight, and 0.9375 with sixteen. These values are a design consequence reproduced by the benchmark, not estimated population rates. Resource-partitioned and dependency-aware alternatives were not run on PostgreSQL.

6.3 Service-layer overhead

Table 3: Descriptive PostgreSQL service-layer benchmark: one worker, history depth 50, 30 repetitions per operation. Values include transaction finalization but exclude HTTP and model inference.

Operation	p50 ms	p95 ms	Sequential ops/s
GST transition	30.967	54.685	29.952
State reconstruction	7.298	12.299	129.688
THSS compilation	61.806	68.889	16.937
Governed-decision reconstruction	10.537	17.011	95.350
Audit lookup	2.122	5.501	399.149
Invalidate + rebuild	52.996	74.336	18.295
Enter-in-error + rebuild	69.314	87.309	13.818

Peak process RSS was approximately 109 MiB. These measurements describe one local, in-process service-layer environment and are not public-HTTP latency, deployment capacity, or production throughput.

6.4 Prospective two-model evaluation

The solver-v5 benchmark completed all 1,152 model-condition cells with zero errors and zero retries. Across cells, accuracy was 93.06% for lifecycle, 96.79% for evidence, 99.22% for timeliness, and 99.83% for escalation; 1,027/1,152 cells (89.15%) were exact on all four axes. On the subject-level primary endpoint, Strict THSS reached 63/64 (98.44%) with Claude and 64/64 (100%) with Gemini. The single remaining Strict-THSS Claude error—a cancelled history predicted as satisfied—was retained without post-hoc repair. The 128 construction-review requests served a separate bounded purpose: two model reviews per subject checked deterministic anchor construction while code retained final authority. All 64 constructions were accepted; candidate-slot F1 and due-window exact accuracy were both 1.0.

Table 4: Ten preregistered subject-level contrasts for Strict THSS. Positive effects favor Strict THSS; Holm adjustment spans the complete ten-test family.

Model	Comparator	Effect (pp)	95% bootstrap CI	Holm result	Discordance
Claude	Naive RAG	+32.81	+21.88 to +43.75	$p = 0.00000954$	21
Claude	LWW	+25.00	+15.63 to +35.94	$p = 0.00027466$	16
Claude	Full history	+18.75	+7.81 to +29.69	$p = 0.01098633$	14
Claude	Long context	+14.06	+6.25 to +23.44	$p = 0.01953125$	9
Claude	BTSA	+4.69	not reported	$p = 1.0$, n.s.	not reported
Gemini	Naive RAG	+25.00	+14.06 to +35.94	$p = 0.00027466$	16
Gemini	LWW	+25.00	+15.63 to +37.50	$p = 0.00027466$	16
Gemini	BTSA	0.00	exact tie	n.s.	exact tie
Gemini	Full history	0.00	exact tie	n.s.	exact tie
Gemini	Long context	0.00	exact tie	n.s.	exact tie

Six contrasts remained Holm-significant, but only 9–21 discordant subjects informed those tests, below the planned 51 informative pairs. Claude versus BTSA was not significant, and Gemini tied BTSA, full history, and long context exactly. The exact sign tests remain valid for observed discordances, but the high tie rate limits effect-size precision and confirmatory strength.

7 Discussion

7.1 What the contract adds

The nearest 2026 systems make clear that freshness checks, transactional commit, and commit-time authorization are no longer sufficient novelty claims by themselves. GLHS therefore makes a narrower claim. THSS records the exact governed health disclosure consumed during inference; a model-derived proposal cites that disclosure rather than only a generic authorization token or current base version; GST then checks the bound state and governance coordinates before persistence. The candidate contribution is this disclosure-conditioned health read–write dependency and its auditability, not the constituent database, provenance, consent, concurrency, transaction, or authorization mechanisms.

7.2 Relation to governed and transactional agent memory

MemTX, MemTxn, Cordon, and TOKI establish strong transaction/write-correctness precedents [13, 14, 21, 22]; CommitGuard and Provenact establish commit-time and policy-state authorization precedents [24, 25]; GateMem, MemClaw, and HealthClaw establish governed/private memory precedents [26, 15, 16]. GLHS therefore narrows the problem to a health-specific governed disclosure that remains relevant when a later model-derived action requests persistent state change. Its scientific value depends on enforcing that provenance-sensitive dependency on model-derived paths and on testing governance-writer races, both of which are explicit limitations of the current evidence.

7.3 Privacy, safety, and operational trade-offs

PII, PHI, and other sensitive attributes are treated as governed data rather than ordinary model context. THSS determines what is disclosed for the actor, purpose, task, consent state, and policy; direct identifiers or sensitive classes may be omitted, minimized, or pseudonymized before model access while provenance remains in the governed record. Evaluation artifacts use synthetic or de-identified inputs where applicable, opaque study-local subject tokens, and content hashes. These measures do not establish legal “minimum necessary” compliance, anonymization, sender authentication, or a general privacy guarantee.

The concurrency experiment exposes a trade-off: a profile-global version counter is simple to reason about but makes unrelated writes compete. At 16 writers the unrelated-slot false-stale rate was 0.9375. Finer-grained versions could reduce contention but require separate safety evaluation because lost updates and cross-dependency errors can reappear when the version boundary is split.

8 Limitations and Versioning Roadmap

The prospective model cohort contains only 64 synthetic subjects, and frequent ties left 9–21 informative pairs in significant comparisons rather than the planned 51. Strict THSS was not significantly better than BTSA for Claude; Gemini tied BTSA, full history, and long context. No independent human clinical adjudication or real-world clinical validation is reported.

The current gateway retains a base-version-only proposal path and does not enforce provenance-sensitive downgrade prevention for every Strict-THSS-derived model proposal after review. State-version commit uses a profile-row lock and rereads consent in the same service transaction, but atomicity against concurrent authorization, consent, or policy writers has not been demonstrated with race tests. No

single baseline is verified to match every governance semantic while differing only in exact snapshot binding. Service/API tests do not include a deployed-boundary attack matrix across HTTP, cache, and retrieval. The performance run used one in-process worker.

For later arXiv replacements, the intended evidence progression is explicit rather than cosmetic: a substantive next version should report newly frozen evidence for mandatory snapshot-bound policy semantics, real database governance-writer race tests, a binding-only matched comparator, independent adjudication, and/or finer-grained concurrency only if those experiments are actually completed. Prior version artifacts and hashes should remain available rather than being silently relabeled.

9 Conclusion

GLHS treats persistent longitudinal health AI as a governed read-write process. THSS records the state and governance context shown to the AI; a snapshot-bound proposal carries those source coordinates forward; GST checks that they remain admissible before persistence. The implementation rejected the tested context mismatches, prevented stale commits in the tested state-version races, and exposed the false-stale cost of coarse profile-global versioning. In a frozen synthetic two-model cohort, Strict THSS reached 98.44% all-axes exact with Claude and 100% with Gemini, with six prespecified Holm-significant comparisons but substantially fewer informative pairs than planned. The evidence supports a traceable software path from governed disclosure to persistent write under the tested conditions, not clinical benefit, universal downgrade resistance, governance-race atomicity, or regulatory compliance.

Statements and Declarations

Competing interests. The authors have no relevant financial or non-financial interests to disclose.

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Ethical approval. Not applicable. The experiments were software evaluations conducted on synthetic data and did not involve human participants, human biological material, or identifiable patient data.

Consent to participate. Not applicable. No human participants were recruited.

Consent to publish. Not applicable. The manuscript contains no identifiable information from individual participants.

Data and code availability. Source code and reproducibility materials are publicly available in the CLARA-Care repository: <https://github.com/Project-CLARA-HBT/CLARA-Care>. The manuscript-level implementation freeze is Git revision 2960acb590fda7ee914b6caded961c89f5e3b8e4; run-specific identifiers are reported above.

Use of generative AI. Generative AI was used only to help restructure and edit author-supplied text. It did not generate the reported data or run the analyses. The authors checked the scientific content and remain responsible for the manuscript.

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